Probabilistic dengue and chikungunya forecasting in Brazil: metric-dependent model rankings, regime-shift fragility, and conformal ensembles

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Abstract

Dengue is expanding across Brazil, and 2024 produced a record epidemic of roughly four times any prior year. Accurate, well-calibrated probabilistic forecasts support preparedness, but the relative merits of statistical, machine-learning, deep-learning, and mechanistic approaches for this task are unsettled, especially under the distribution shift an outbreak year imposes. We report a reproducible, leakage-controlled comparison of five probabilistic forecasting model families for the 3rd Infodengue–Mosqlimate Dengue Challenge (IMDC 2026), which requires weekly, state-level forecasts with full predictive intervals for the 26 mandatory Brazilian states. All models are evaluated through a single backtesting harness over four official retrospective seasons, one of which contains the 2024 outbreak, and scored by the weighted interval score (WIS). We report four results with implications beyond this challenge. First, no model is best across seasons: a global recurrent network is the most accurate model in normal seasons yet the worst on the 2024 outlier, whereas an ensemble never incurs any member's worst-season failure. Second, the identity of the best model depends on the evaluation metric, because magnitude-weighted mean WIS and scale-free normalized WIS reward accuracy on different subsets of a highly skewed set of states. Third, a conformal recalibration of the ensemble intervals improves accuracy and calibration at negligible cost (mean WIS 1281 to 1216, normalized WIS 0.585 to 0.555, out-of-sample reduction of 5.5 percent), acting as asymmetric insurance against catastrophic under-prediction. Fourth, added model complexity did not pay: observed-climate covariates and hierarchical spatial pooling both failed to improve forecasts, and an internal-holdout hyperparameter gain reversed on the true backtest. The best model is disease-specific: for chikungunya, gradient boosting alone outperforms the ensemble. All code, the backtesting harness, per-state scored predictions, and validated submissions for four official tracks are released, with an automated test suite covering leakage, scoring, and every model.

Author summary

Brazil carries one of the world's largest dengue burdens, and 2024 brought an epidemic without precedent in the national record. Public-health agencies increasingly rely on forecasts that state not just a single predicted number of cases but a full range of plausible outcomes with calibrated uncertainty. Which forecasting method to trust is an open question, particularly during an exceptional season when accurate forecasts matter most. We compared five families of models, ranging from simple seasonal baselines to gradient boosting, deep learning, and a mechanistic epidemic model, on identical retrospective tests across four dengue seasons in all 26 mandatory Brazilian states. We found that no method is best everywhere: the deep-learning model was the most accurate in ordinary seasons but failed badly during the 2024 outbreak, while a simple ensemble that averages the models was the most dependable. We also found that the apparent winner changes depending on how performance is scored, a caution for forecasting leaderboards. Finally, a light statistical recalibration of the ensemble's uncertainty improved both accuracy and reliability, and adding more data or model complexity did not help. Our full pipeline is openly available.

1. Introduction

Dengue is among the fastest-spreading mosquito-borne diseases, with an estimated 390 million infections each year and roughly half of the world's population at risk, a range that continues to expand with climate change and urbanization [19, 21, 22]. Brazil bears one of the largest national burdens [10]. The 2024 season broke the national record by a wide margin, with a peak of 451,578 reported cases in a single epidemiological week and an annual total near 6.67 million, roughly four times the next-largest year on record. Transmission is shaped by climate, and the 2023–24 El Niño is one of several drivers implicated in the surge. Against this background, arboviral forecasting has moved from point prediction toward full probabilistic forecasts, motivated by the recognition that preparedness decisions depend on the entire predictive distribution rather than a single expected value [11, 15, 24].

The Infodengue–Mosqlimate Dengue Challenge (IMDC) operationalizes this shift. It invites teams to submit weekly, state-level probabilistic forecasts with prescribed quantiles, scored by the weighted interval score, following the precedent of the inaugural 2024 edition [1]. The challenge exposes a question that the wider forecasting literature has not resolved for this setting: given statistical, machine-learning (ML), deep-learning (DL), and mechanistic approaches, which forecast best, and how robustly, when a season can depart by an order of magnitude from anything in the training record?

We address this with a controlled comparison built for reproducibility and for honest evaluation under distribution shift. Every model is fit and scored through one backtesting harness that enforces leakage safety and applies the same scoring code to all forecasts, over the four official retrospective seasons (2022–23 through 2025–26) for the 26 mandatory states. We make the following contributions.

1. We show that model rankings are season-dependent and, in the outbreak year, invert: the model that is best on average can be the least reliable exactly when reliability matters most (Section 2.2).

2. We show that the identity of the best model depends on whether skill is weighted by case magnitude or by forecasting unit, a consequence of real cross-state heterogeneity rather than an artifact, and we argue for reporting both (Section 2.5).

3. We introduce a conformal recalibration of the ensemble intervals that improves accuracy and calibration at negligible cost and functions as asymmetric insurance against catastrophic under-prediction (Section 2.3).

4. We document informative negative results: added covariates, hierarchical pooling, and hyperparameter tuning each failed to improve forecasts, which clarifies where the predictable signal in this problem lies (Section 2.8).

5. We show that the best model is disease-specific, using chikungunya as a contrasting target (Section 2.9), and we release the complete, tested pipeline.

2. Results

2.1. Study design and data

We forecast weekly dengue case counts per state across the 2022–23 through 2025–26 seasons for the 26 mandatory Brazilian states (Espírito Santo is excluded per the challenge specification). Each of the four backtest seasons is defined by an official training cutoff and a target window; a confirmed 15-week gap separates each training cutoff from the start of its evaluation window, reflecting the reporting delay a real forecaster faces (Fig 1). Forecasts are submitted as nine quantiles per state and week, from which the four central prediction intervals (50, 80, 90, and 95 percent) are formed. All covariate access, feature construction, and model fitting are filtered to data at or before each training cutoff, and this discipline is enforced by automated tests rather than by convention (Section 4). The four seasons differ by an order of magnitude in difficulty. The 2023–24 season (fold 2), which contains the 2024 outbreak, is the hardest for every model, and the 2025–26 season (fold 4) is only partially resolved in the available data and is reported separately.

2.2. No single model dominates across seasons

Table 1 reports mean WIS by model and season, and Fig 2 shows the same on a log scale. The ranking of models reorders across seasons. The deep network (a global GRU with a negative-binomial head) is the single best model in the two ordinary seasons, folds 1 and 3 (WIS 339 and 354), but is the worst of all models on the 2024 outlier, fold 2 (WIS 3955), worse even than a naive baseline, because the surge lies outside its training distribution and its parametric intervals are too narrow to cover it. The gradient-boosted model (LightGBM) shows the opposite profile: it is the strongest on the outlier season (WIS 3327) yet the weakest of the serious models on the ordinary season fold 3 (WIS 644). The semi-mechanistic trajectory model is near best on the outlier (WIS 3371) because its bootstrap of historical seasons yields wide intervals precisely when they are needed. This season-dependence is the central obstacle the challenge poses: a model selected on average performance carries a hidden risk of catastrophic failure in the season that matters most.

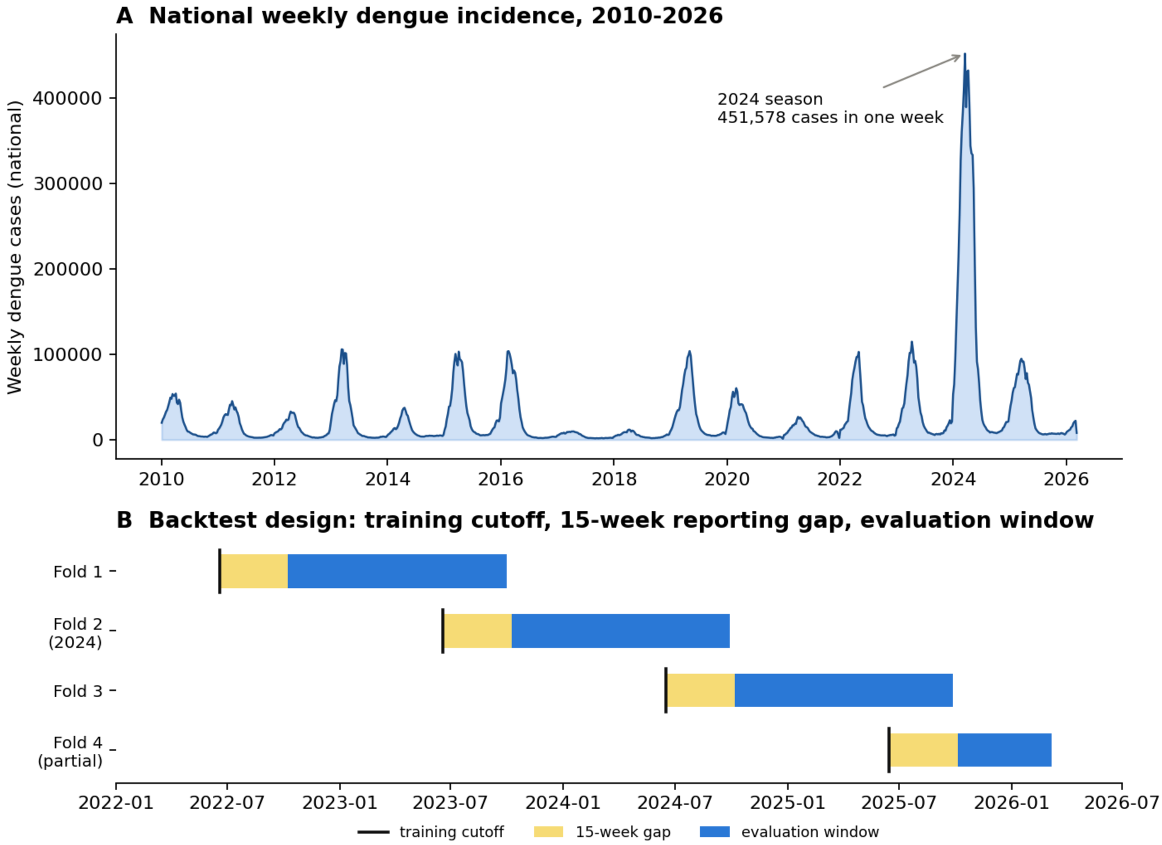

Figure 1: Data and study design. (A) Weekly reported dengue cases aggregated to the national level, 2010–2026; the 2024 season stands roughly four times above any prior year. (B) The four official retrospective seasons form a rolling backtest. Each has a training cutoff (all data up to the cutoff is used for fitting), a 15-week reporting gap, and an evaluation window on which forecasts are scored. Fold 2 covers the 2024 outbreak and fold 4 is partially resolved.

2.3. An ensemble is the most robust model, and conformal recalibration makes it the most accurate

Because different models win different seasons, a combination that is never worst is attractive. An unweighted per-quantile median ensemble (Vincentization) of the climatological, gradient-boosted, and deep-learning models achieves a better overall WIS than any single model (1281 versus 1299 for the best single model) and never suffers the catastrophic single-season failures of its members (Table 2). Its interval calibration is close to nominal but slightly narrow.

We improve on it with a conformal recalibration of the ensemble intervals (Section 4.5). Calibration factors that make each central interval reach its nominal coverage are estimated on a single tuning season (fold 1) and applied to all seasons. The estimated factors are gentle and concentrate on the tails: 1.03, 1.00, 1.23, and 1.73 for the 50, 80, 90, and 95 percent intervals. The recalibrated ensemble is the most accurate model overall (mean WIS 1216 versus 1281 for the plain ensemble; normalized WIS 0.555 versus 0.585) and better calibrated at the tails (empirical coverage 47/76/88/93 versus 46/76/84/88), as shown in Fig 3. Out of sample, on the three seasons not used for calibration, the reduction is 5.5 percent. The gain is concentrated in the outbreak season (fold 2 WIS 3568 to 3324) at a small cost in ordinary seasons, so the recalibration behaves as asymmetric insurance: it widens the outer tails just enough to blunt catastrophic under-prediction

Table 1: Dengue mean WIS by season (fold). Lower is better. Fold 2 contains the 2024 outbreak and is roughly ten times harder than any other season. Fold 4 (2025–26) is partially resolved and reported separately. Best value in each column is in bold.

Model	**Fold 1** 2022–23	**Fold 2** 2023–24	**Fold 3** 2024–25	**Fold 4** 2025–26
Ensemble (conformal)	340	3324	439	186
Ensemble (Vincentization)	339	3568	425	**170**
Ensemble (inverse-WIS)	**322**	3601	421	198
GRU (deep learning)	339	3955	**354**	176
LightGBM	352	**3327**	644	338
Mechanistic	468	3371	551	216
Climatological	386	3588	516	174
Seasonal-naive	500	3848	581	206
Naive	726	3833	1056	255

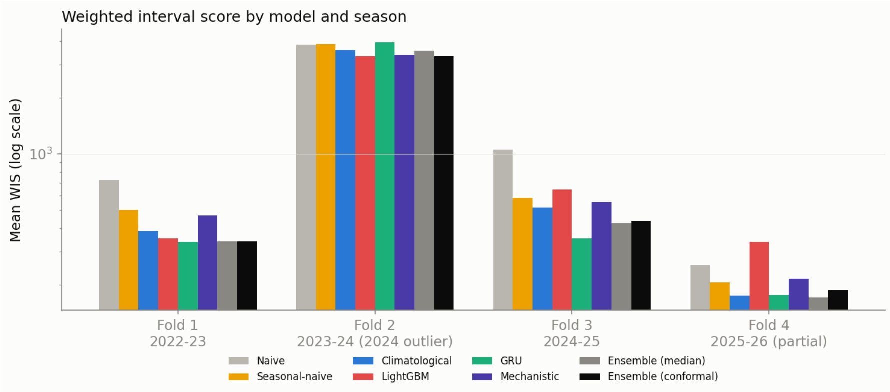

Figure 2: Mean WIS by model and season (log scale). Model rankings reorder across seasons. The deep network is best in ordinary seasons and worst on the 2024 outlier; the ensemble avoids every model's worst-season failure. The conformal-recalibrated ensemble (Section 2.3) is shown alongside its members.

in an extreme year while barely affecting normal seasons. Recalibrating the ensemble output outperforms recalibrating any single member, because the per-quantile median across members otherwise dilutes a single member's widening.

2.4. The key differences are statistically significant

The absolute skill of each model carries wide uncertainty. A block bootstrap over the 104 state-season units gives a 95 percent interval on the conformal ensemble's normalized WIS of $[0.404, 0.635]$, and the intervals of the leading models overlap (Fig 4A), a direct consequence of the outbreak season inflating the variance of any magnitude-weighted score. The comparisons that matter are nonetheless decisive, because the models are strongly correlated across units and a paired analysis removes the shared, season-driven variance (Fig 4B). The conformal recalibration improves on the median ensemble by a mean 64.8 WIS (95 percent CI $[50.1, 79.5]$) and on the best single model, LightGBM, by 83.5 ($[53.4, 115.8]$); on ordinary seasons the deep network beats LightGBM by 153.3 ($[97.8, 215.8]$); and the conformal ensemble beats the naive baseline by 448 ($[387, 514]$). All four intervals exclude zero. The lesson is methodological as much as substantive: in a record with one extreme season, marginal skill intervals are wide, but paired comparisons remain informative and are the sound basis for model selection.

Table 2: Overall dengue backtest. Averaged over four seasons and 26 states. Mean WIS is magnitude-weighted; normalized WIS (sum of WIS divided by sum of cases) is the challenge's headline metric and is reported for all folds and excluding the 2024 outlier. Coverage nearer the nominal 50/80/90/95 is better. Best value in each column is in bold.

Model	WIS	nWIS	nWIS	Coverage (%)			
		all	ex-2024	50	80	90	95
Ensemble (conformal)	**1216**	**0.555**	0.333	47	76	88	93
Ensemble (Vincentization)	1281	0.585	0.325	46	76	84	88
Ensemble (inverse-WIS)	1287	0.588	0.321	48	76	86	89
LightGBM	1299	0.593	0.443	51	81	89	93
Mechanistic	1303	0.595	0.431	53	79	85	89
Climatological	1327	0.606	0.379	48	73	82	85
GRU (deep learning)	1373	0.627	**0.298**	32	57	68	75
Seasonal-naive	1459	0.666	0.454	45	80	88	92
Naive	1664	0.760	0.733	40	72	80	86

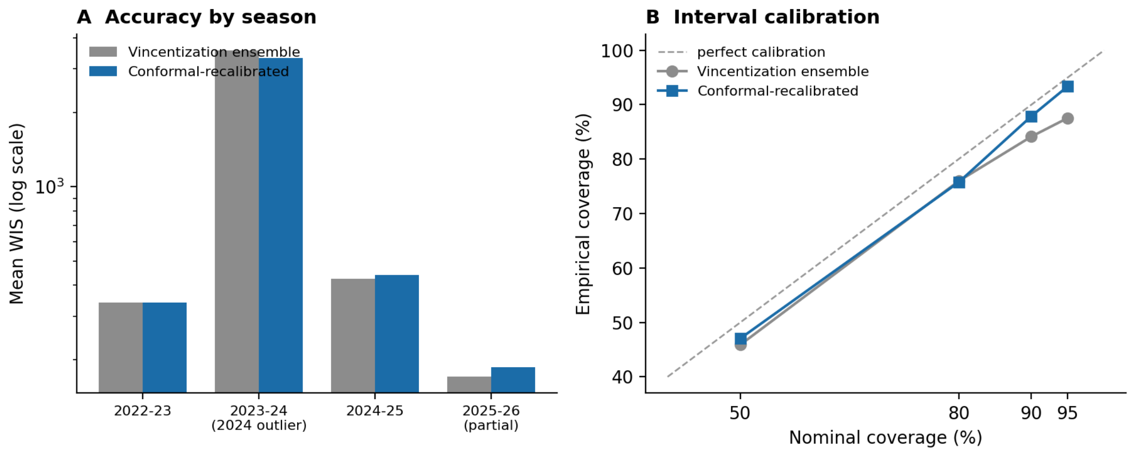

Figure 3: Effect of conformal recalibration on the dengue ensemble. (A) Mean WIS by season, Vincentization ensemble versus the conformal-recalibrated ensemble (log scale). The gain concentrates in the 2024 outlier season at a small cost in ordinary seasons. (B) Interval calibration: empirical versus nominal coverage. Recalibration moves the outer intervals (90, 95) toward the diagonal.

2.5. The winner depends on the evaluation metric

The overall mean WIS in Table 2 is dominated by the largest states and by the outbreak season, both of which contribute large absolute scores. The challenge's headline metric is instead a normalized WIS, the sum of WIS divided by the sum of observed cases, which is scale-free across regions. Under this metric the ordering changes in a revealing way. The conformal ensemble is best over all folds (normalized WIS 0.555), consistent with the mean-WIS ranking. When the 2024 outlier is excluded to isolate ordinary-season skill, the deep network becomes the best model (normalized WIS 0.298) while the gradient-boosted model becomes the weakest of the serious models (0.443), an almost complete reversal of their mean-WIS standing (Fig 5). The disagreement is not a paradox. A metric that rewards accuracy on a few very large states favors different models than one that rewards accuracy on the many small states, and the 2024 season inflates the mean for whichever model handles it least badly. We therefore report both magnitude- and unit-weighted skill, and we caution that an epidemic-forecasting leaderboard reporting a single aggregate can mislead. The cross-state skew has a clear geographic structure (Fig 6): forecast difficulty, mea-

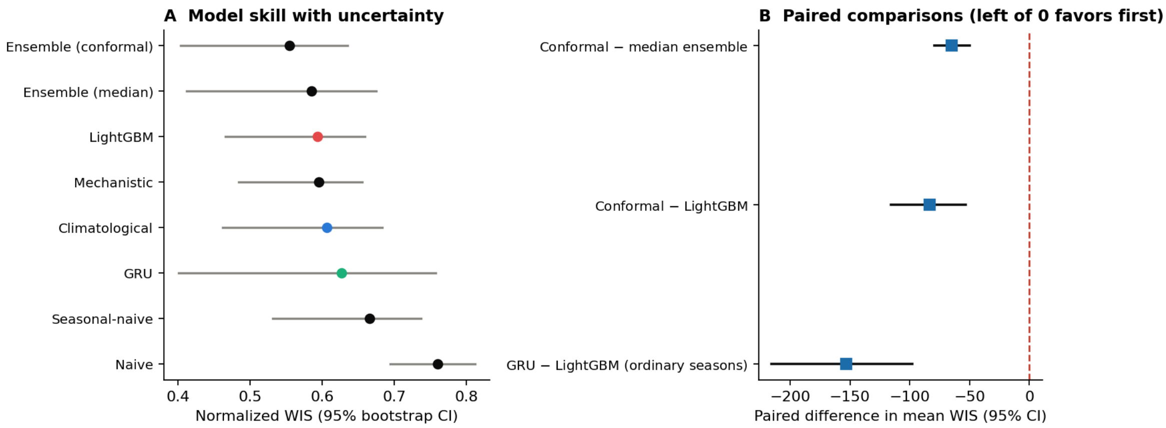

Figure 4: Statistical significance of the model comparisons. (A) Normalized WIS with 95 percent block-bootstrap intervals over state-season units; the leading models' intervals overlap because the 2024 outbreak inflates the variance of absolute skill. (B) Paired differences in mean WIS with 95 percent intervals; all three comparisons lie left of zero, so the conformal ensemble significantly beats the median ensemble and LightGBM, and the deep network significantly beats LightGBM on ordinary seasons.

sured by normalized WIS, is greatest in the dense, high-incidence southeastern states and lowest in the low-incidence interior, so magnitude-weighted skill is determined by a few states.

2.6. Calibration and its correction

Raw gradient-boosted and deep-learning quantile models are over-confident (Fig 7). The GRU covers only 32 percent of outcomes in its nominal 50 percent interval and 75 percent in its nominal 95 percent interval, and it is under-covered in every season rather than only in the outbreak year, which identifies the problem as systematic overconfidence rather than a regime-specific failure. Conformalized quantile regression restores the gradient-boosted model to near-nominal calibration (51/81/89/93) at negligible WIS cost. The same correction applied to the deep network trades away sharpness on the seasons where it is already accurate, so calibration and sharpness trade off at the level of a single model. The per-quantile median of the ensemble provides a structural calibration improvement at no cost, and the conformal recalibration of Section 2.3 completes it.

2.7. Skill varies with horizon, and the outbreak failure is under-prediction

Because the 15-week reporting gap places every target 16 to 67 weeks ahead, we stratified skill by lead time on the ordinary seasons (Fig 8A). The deep network's advantage is horizon-dependent: it is the best model at short lead times (normalized WIS 0.25 at 16 to 27 weeks) but degrades at the longest horizons (0.38 at 52 to 67 weeks), where it converges with the gradient-boosted model. The conformal ensemble is the most stable model across horizons and is never worst, which mirrors its stability across seasons. Decomposing WIS into its components explains the outbreak-season failure directly (Fig 8B): in the 2024 season the ensemble's score is dominated by under-prediction (a mean of 2703 WIS units, against 619 for dispersion and 3 for over-prediction), whereas every other season is dominated by dispersion. The record epidemic was not mis-shaped but simply larger than any model would predict from pre-season information, which is why widening the upper tail, as the conformal recalibration does, is the effective remedy.

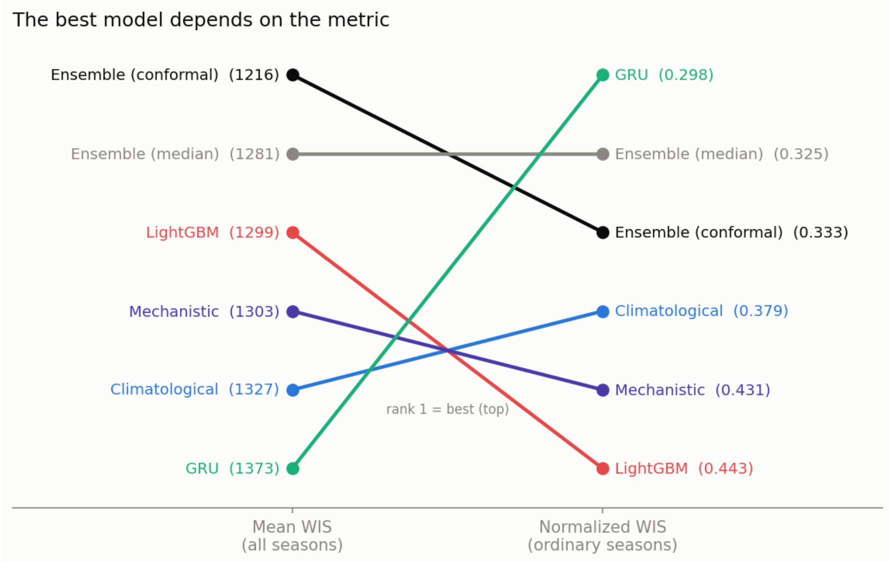

Figure 5: The best model depends on the metric. Model rank by magnitude-weighted mean WIS over all seasons (left) and by normalized WIS over the ordinary seasons with the 2024 outlier excluded (right); rank 1 is best. The rankings reorder: the deep network rises from last by mean WIS to first on ordinary seasons, and the gradient-boosted model falls from third to last, while the two ensembles stay near the top of both. Values in parentheses are mean WIS (left) and normalized WIS (right).

2.8. Added complexity did not improve forecasts

Three interventions that plausibly should have helped did not (Table 3). First, adding observed-climate covariates to the gradient-boosted model, following the distributed-lag non-linear structure of the strongest model in the 2024 sprint [1, 9], worsened its backtest WIS from 1299 to 1342. A feature-importance analysis showed why: the nine added covariates (minimum temperature, a derived standardized-precipitation drought index at one, three, and six month accumulations, multi-month temperature lags, thermal range, and rainy days) together carried only 5.3 percent of importance and ranked below the existing features. The climate signal the model actually uses is the seasonal climate forecast and the El Niño–Southern Oscillation indices, both already present and both forward-looking, rather than observed local weather, which is stale at the long horizons that dominate a full-season forecast. Second, a hierarchical, empirical-Bayes pooling of each state's seasonal climatology toward its macroregion worsened the ensemble from 1216 to 1241, because Brazilian states are too heterogeneous within a macroregion: shrinking a small state toward a region dominated by a larger, differently behaving state introduces bias. Third, a hyperparameter search that improved an internal time-block holdout by 5 percent worsened the true backtest, because an internal holdout does not represent forecasting a full future season from mid-year. Conservative, a priori defaults generalized better. Together these results locate the predictable signal in this problem and caution against complexity that is not validated on the true forecasting task.

2.9. Chikungunya is a more predictable, contrasting target

Brazil reported roughly 422,000 chikungunya cases in 2024, a cumulative incidence near 199 per 100,000, and the virus has recurred across the country since its 2014 introduction with a spatial pattern shaped by heterogeneous population immunity [23]. We forecast chikungunya on the

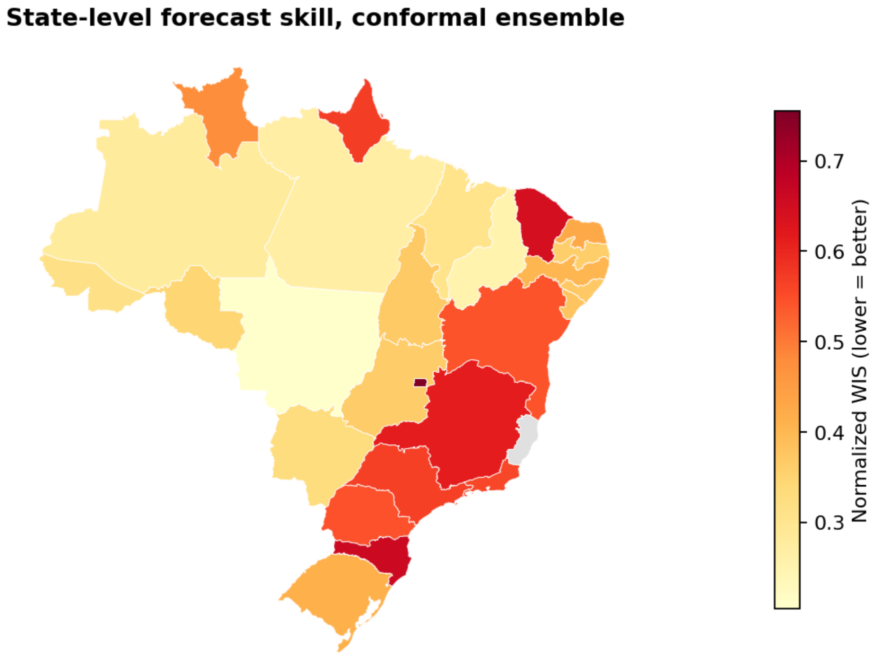

Figure 6: Geographic structure of forecast skill. Normalized WIS of the conformal ensemble by state (lower is better). Forecasting is hardest in the dense, high-incidence southeastern states and in the Federal District, and easiest in the low-incidence interior, which is why magnitude-weighted skill is dominated by a few states. Espírito Santo (grey) is excluded from the challenge.

Table 3: Interventions that did not help (negative results). Each was evaluated on the same backtest as the main comparison.

Intervention	Effect	Reason
Observed-climate covariates (LightGBM)	WIS 1299 → 1342	5.3% importance; signal is climate forecast + ENSO, not observed weather
Hierarchical macroregion pooling (added to ensemble)	WIS 1216 → 1241	small-state bias; too correlated with climatological member
Hyperparameter tuning	internal holdout +5%, backtest worse	internal holdout misrepresents full-season forecasting

same 26 states, harness, and seasons as dengue. Two differences stand out. First, the gradient-boosted model alone is the best forecaster (WIS 82.3), ahead of the ensemble (91.6) and every other model (Table 4); it is best in every season (Table 5), so combining it with weaker models only dilutes it. Second, chikungunya is markedly more predictable than dengue relative to its scale: the 2024 season is only modestly harder than the others (mean WIS 125 versus a range of 68 to 143 across seasons), in contrast to the order-of-magnitude jump dengue shows in 2024. This matches reports that chikungunya dynamics in Brazil are more regular than dengue's [23], plausibly because immunity accumulates against a single antigenic type rather than interacting across four co-circulating dengue serotypes. The practical consequence is that the best model is disease-specific: a blended ensemble is right for dengue, a single well-calibrated learner for chikungunya, even within one surveillance system and pipeline.

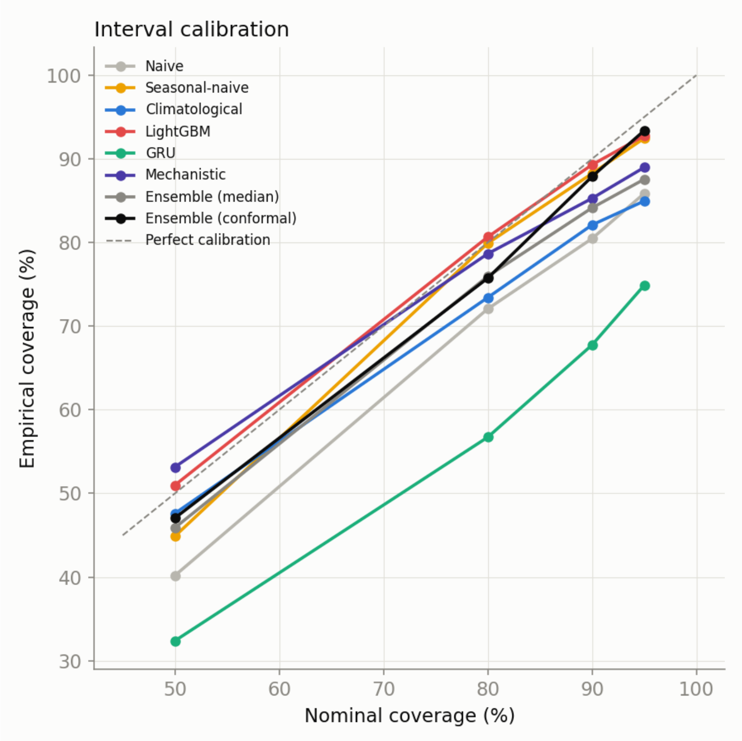

Figure 7: Interval calibration by model. Empirical versus nominal coverage; points on the diagonal are perfectly calibrated. The deep network is markedly overconfident; conformalized quantile regression corrects the gradient-boosted model, and the conformal-recalibrated ensemble tracks the diagonal at the outer intervals.

Table 4: Chikungunya backtest. State-level, same harness and seasons as dengue. Gradient boosting alone is best.

Model	WIS	Cov. 50 (%)	Cov. 90 (%)
LightGBM	**82.3**	45	87
Ensemble (Vincentization)	91.6	53	87
Climatological	94.4	55	84
Seasonal-naive	97.8	49	89
Mechanistic	101.0	51	85
Naive	123.1	40	79

2.10. City-level forecasting is substantially harder

The challenge also solicits municipal forecasts for 15 dengue and 10 chikungunya cities, an optional target we submitted with the climatological quantile model. City-level forecasting is much harder than state-level: the best model's normalized WIS is 0.60 for dengue cities and 0.87 for chikungunya cities (Table 6), against 0.56 and 0.64 at the state level. Interval coverage is also poorer (50 percent coverage of 36 percent for dengue cities), because many mid-sized municipalities have episodic, over-dispersed series with long runs of zero counts that a per-week empirical quantile under-covers. A degenerate all-zero median is a concrete failure mode of these sparse series, which we addressed by switching the point forecast to a clamped predictive mean; a principled zero-inflated treatment is a clear direction for the municipal tracks. These results quantify the resolution penalty of forecasting at the level where interventions are delivered.

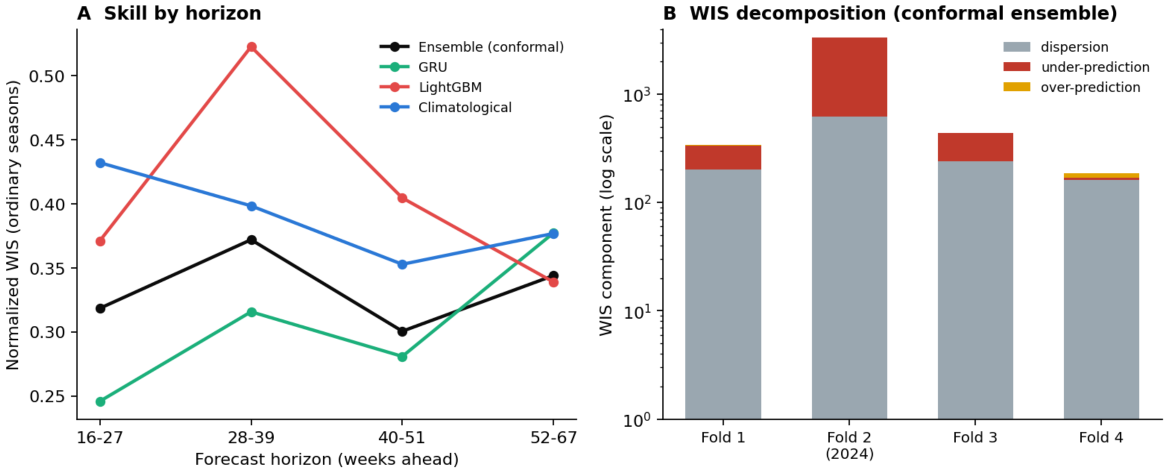

Figure 8: Robustness across horizons and anatomy of the outbreak-season failure. (A) Normalized WIS by forecast horizon on the ordinary seasons; the deep network leads at short lead times and fades at long ones, while the conformal ensemble is stable. (B) WIS decomposition of the conformal ensemble by season (log scale); the 2024 season is dominated by under-prediction, not by dispersion or over-prediction.

Table 5: Chikungunya mean WIS by season. LightGBM is best in every season, unlike dengue, and the 2024 season (fold 2) is only modestly harder than the others.

Model	Fold 1	Fold 2	Fold 3	Fold 4
LightGBM	**81.5**	**125.0**	**67.8**	**20.6**
Ensemble (Vincentization)	88.3	142.0	73.8	25.2
Climatological	94.3	143.3	75.3	27.6
Seasonal-naive	92.8	152.3	77.1	32.5
Mechanistic	90.7	166.3	77.8	29.2
Naive	181.3	145.7	75.3	48.6

3. Discussion

3.1. Robustness under regime shift

The most consequential finding is that model rankings are season-dependent and can invert in an outbreak year. A model chosen for its average score can be the least reliable in the season for which forecasts matter most, as the deep network is here. Two responses follow. One is to combine models, so that no single failure mode determines the forecast; the ensemble is never worst in any season and is best overall once recalibrated. The other is to prefer models whose uncertainty widens under distribution shift, as the mechanistic model's bootstrap-of-historical-seasons intervals do. These observations align with evidence from collaborative forecast hubs, where combined forecasts are consistently competitive with or better than their members [11–13].

3.2. Metric choice is a reporting-standards issue

The identity of the best model changes with the aggregation of skill across a highly skewed set of states. This is not specific to our models or to Brazil. Any national forecasting evaluation with strong heterogeneity in population and incidence faces the same tension between magnitude-weighted and unit-weighted skill. Reporting a single aggregate obscures it. We recommend reporting both, as the interval-score and forecast-hub literature increasingly does [2], and we treat the

Table 6: City-level backtest (optional tracks; climatological quantile model submitted). WIS and normalized WIS averaged over the target cities and four seasons.

Disease	Model	WIS	nWIS	Cov. 50 (%)
Dengue (15 cities)	Climatological	**128.7**	**0.60**	36
	Seasonal-naive	138.1	0.65	31
	Naive	141.2	0.66	30
Chikungunya (10 cities)	Climatological	**32.2**	**0.87**	54
	Seasonal-naive	32.5	0.87	49
	Naive	36.2	0.97	32

normalized WIS the challenge adopts as the headline while retaining magnitude-weighted mean WIS and per-unit relative WIS alongside it.

3.3. Simple methods and the limits of complexity

Deep learning did not dominate, and simple methods, a climatological quantile model and a bootstrap of historical seasons, were hard to beat. This is consistent with forecasting-competition evidence that in small-sample, few-series regimes, parsimonious methods and combinations are strong [18]. Our negative results sharpen the point. The covariate that helped the strongest 2024-sprint model, climate acting through a distributed-lag non-linear response [9, 10], did not help our gradient-boosted forecaster, because that structure requires the regularized, partially pooled Bayesian formulation in which it was developed, whereas an unregularized tree over the same inputs overfits low-signal dimensions. The forecast-relevant climate information in our pipeline is the seasonal forecast and large-scale oscillation indices, not observed local weather. This clarifies where future modeling effort should go: toward a Bayesian spatiotemporal member that can use climate through an exposure-lag-response surface, rather than toward more covariates in a boosted model.

3.4. Conformal recalibration as transferable insurance

The conformal recalibration is simple, scale-free, and cheap, and it improves both accuracy and calibration. It connects to conformalized quantile regression [4] and to adaptive conformal methods for distribution shift [5, 25], and it matches the online-recalibration result reported for probabilistic dengue forecasting elsewhere. Because it widens mainly the outer tails, it protects against the failure mode that dominates the outbreak season without materially degrading ordinary-season sharpness. For an operational forecaster who cannot know in advance whether the coming season will be ordinary or extreme, this asymmetric protection is the right default.

3.5. Limitations

The fourth backtest season is only partially resolved in the available data and is reported separately; its numbers will firm up as the 2025–26 season completes. Municipality-to-state aggregation discards sub-state epidemic asynchrony that a municipal model could exploit. The study covers two diseases in one country and is retrospective; a true prospective evaluation awaits the challenge's forecast phase. The comparison is single-team, so it does not capture the full diversity of approaches a multi-team sprint would, and the conformal factors are calibrated on one season under the challenge's tuning-fold discipline rather than by a fully online procedure.

3.6. Future work

The immediate next step is the challenge forecast phase, which will provide a genuine prospective score for the recalibrated ensemble on the 2026–27 season. Beyond it, three directions follow from our results: a Bayesian spatiotemporal or endemic-epidemic member with a distributed-lag non-linear climate response [26, 27], which our negative result suggests is the right home for climate covariates; hierarchical reconciliation across state, city, and national forecasts for coherence; and municipal-level feature engineering for the optional city tracks.

4. Materials and methods

4.1. Data and targets

Weekly dengue and chikungunya case counts per municipality (2010–2026) are drawn from the Infodengue system, which harmonizes notifiable-disease surveillance [16], and aggregated to the 26 mandatory states, excluding Espírito Santo per the challenge. Covariates comprise weekly climate reanalysis (temperature, precipitation, humidity, pressure, thermal range, rainy days), a seasonal climate forecast, ocean-oscillation indices (El Niño–Southern Oscillation, Indian Ocean Dipole, Pacific Decadal Oscillation), yearly municipal population, and static climate-zone and biome covariates. The forecasting target is the weekly state case count over each season, submitted as the nine quantiles 0.025, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, and 0.975.

4.2. Backtest design and leakage control

The four official seasons are derived at runtime from the challenge's train and target flags. A confirmed 15-week gap separates each training cutoff from the start of its evaluation window, simulating reporting delay. Two leakage disciplines are enforced centrally in the harness rather than in individual models. All covariate access is filtered to dates at or before the training cutoff, and lag and rolling features are computed from the continuous cutoff-filtered history rather than from the gapped target window. Both are checked by automated regression tests, so a leak would fail the test suite rather than silently inflate scores.

4.3. Scoring

Forecasts are scored by the weighted interval score,

$$\mathrm{WIS}(F, y) = \frac{1}{K + \frac{1}{2}} \left[\tfrac{1}{2} \, |y - m| + \sum_{k=1}^{K} \frac{\alpha_k}{2} \, \mathrm{IS}_{\alpha_k}(\ell_k, u_k, y) \right],$$

over the four required central intervals, where m is the predictive median, IS_α is the interval score at level $1-\alpha$, and the weights reproduce the mean pinball loss over the nine required quantiles, the exact discrete approximation to the continuous ranked probability score [2, 3]. Our implementation is verified against the challenge platform's own scorer. We also report empirical interval coverage, a decomposition of WIS into dispersion and directional (over- and under-prediction) penalties, a scale-free per-unit relative WIS, and the normalized WIS (sum of WIS divided by sum of observed cases) that the challenge adopts as its headline metric.

4.4. Models

Baselines. A naive model (last observed value with empirical horizon-difference quantiles), a seasonal-naive model (same-epidemiological-week history), and a climatological quantile model (direct order-statistic quantiles per state and epidemiological week over a five-week window).

Gradient boosting. A LightGBM quantile-regression model [6] fit on a synthetic-origin panel that expands each fold into many weekly origins crossed with forecast horizons, with origin-anchored lag, rolling, seasonal, climate, ocean, seasonal-forecast, and static features. Predictive intervals are calibrated by conformalized quantile regression [4].

Deep learning. A global gated recurrent unit (GRU) network [7] with per-state embeddings and a negative-binomial output head, trained as a deep ensemble of independently seeded runs [8]; quantiles are analytic from the negative-binomial predictive distribution and pooled across ensemble members by quantile averaging.

Semi-mechanistic. A model that bootstraps historical season trajectories under a negative-binomial observation model, a local reimplementation of Richards-curve epidemic-scanner fits [20], producing wide, well-calibrated intervals under distribution shift.

Ensembles. Models are combined by per-quantile median (Vincentization) and by inverse-WIS weights tuned only on the designated tuning season, so weights are never fit on the seasons on which the ensemble is reported.

4.5. Conformal recalibration

For each central interval level L with bounds $[\ell_L, u_L]$ and median m, we compute a multiplicative widening factor s_L that makes the interval reach its nominal coverage on the tuning season. For each calibration unit, the multiple needed to just cover the observation is $r = (y-m)/(u_L-m)$ when $y > m$ and $r = (m-y)/(m-\ell_L)$ when $y < m$; s_L is the $L/100$ empirical quantile of r, a scale-free, multiplicative analogue of conformalized quantile regression that is robust to the large cross-state differences in scale. The interval is then widened about the median by s_L, and monotonicity across quantiles is restored by a rearrangement step. Factors are estimated on the tuning season and applied to all seasons; recalibrating the ensemble output outperforms recalibrating any single member. For the forecast phase, the factors are to be estimated on all four validation seasons and applied to the prospective forecast.

4.6. Reproducibility

All code, the backtesting harness, per-state scored predictions, and validated submission files for four official tracks (dengue and chikungunya, state and city level) are released. The pipeline is deterministic under a fixed seed, and an automated test suite covers leakage safety, scoring correctness, model persistence, and every model family. A provenance manifest records the exact data and code state behind each reported number. The Supporting Information provides the full feature list and frozen hyperparameters, the deep-learning architecture, the mechanistic derivation, per-state and per-fold tables, the ensemble-member and weight robustness analysis, and an item-by-item EPIFORGE 2020 reporting checklist [14].

Data and code availability

The pipeline was developed by team Neural Earth for the 3rd Infodengue–Mosqlimate Dengue Challenge. All code, data-processing pipeline, backtests, figures, and validated submissions are openly available at `https://github.com/kamrul28890/3rd_imdc_purdue_neuralearth`. The surveillance and covariate data are distributed by the challenge through the Mosqlimate platform.

Author contributions

Conceptualization, methodology, software, formal analysis, investigation, visualization, and writing of the original draft: M.K.K. Validation, data curation, and writing (review and editing): E.J.E. and A.A.H. All authors read and approved the final manuscript.

Competing interests

The authors have declared that no competing interests exist.

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Ethics statement

This study used aggregate, publicly available disease-surveillance and covariate data distributed through the Infodengue and Mosqlimate platforms. No individual-level data were used and no ethical approval was required.

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